

Round 4 Report — Resolution of Literature Gap Audit (R3-C1...C9)

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Document: Round 4 Progress Report (R3-C1...C9 Complete)

Audit Cycle: Literature Gap Audit (Resolution)

VERIFIED PASS

All numbers re-checked on 2026-10-06: full check suite passes **210/210 checks** (+26/26 data-provenance checks, max|leakage| = 0.00e+00). Every headline number was re-read from result CSVs before inclusion.

EXECUTIVE SUMMARY

Section 0 — Headline Findings & Literature Gap Resolution

All nine items (C1-C9) from the Literature Gap Audit are addressed. On identical features and model, the split protocol alone moves R2 from -1.03 to +0.99. Removing capacity proxies moves it from 0.93 to 0.49; adding back two safe charge-segment features recovers it to 0.81. The AE did not add value on either modality, so it is reported as a negative result and the pipeline proceeds without it. Cross-dataset transfer to CALCE CS2 fails; a within-CS2 control shows the cause is domain shift, not data or feature quality. All 13 audit papers were verified from their full PDFs, and the two anchor numbers of the audit match the originals. One question remains: the thesis title phrasing (Section 10).

1. C1 - Evaluation protocol fixes

- `refit_on_3`: after inner selection, the final model is re-fitted on all three training batteries. Mean per-fold R2 (TD-All): 0.848 -> 0.923; median 0.789 -> 0.950. (Source: `inner_loop_lobo3_results.csv`, re-aggregated 2026-10-06.)
- Reporting standard now applied in all tables: per-fold mean R2 and pooled R2, MAE and RMSE in %SOH, MAPE, and RMSE in Ah. (Source: `phase1_pooled_metrics.csv` - pooled TD-All refit 0.936, MAE 2.20 %SOH, MAPE 2.71 %.)

2. C2 - Protocol-gap experiment (becomes Ch.4 section 4.2)

Same features, same model, only the split changes (verified: `protocol_gap_verify.py`, 23/23):

Split	Oracle-proxy	TD-All	TD-Proxy-Free
Random, same battery	0.992	0.979	0.978
Random, pooled	0.815	0.989	0.978
Nested LOBO, unseen battery	0.663	0.927	0.854
Chronological, same battery	0.179	-1.028	-5.234

- The R3-C2 prediction (random split above 0.97) is met: 0.979.
- Pooling inflates R2. The chronological split scores -1.028 per fold but +0.850 pooled. This is audit factor 5, shown on a fixed representation.
- The LOBO numbers above come from the phase-1 audit run (`refit_on_3`, 3 seeds). The phase-2 protocol-gap experiment (pre-refit stage) produced its own LOBO numbers (0.634/0.891/0.680) on a different row set; Ch.4 quotes them separately with a note, and the two are not compared to each other.

3. C3 - Feature and proxy audit

- `dis_duration` is a direct proxy. In CC discharge, capacity = $I \times t$, so the feature carries the label. It was removed to form the Proxy-Free set.
- `dis_mean_V` is collinear with it (within-battery $r = 0.941$) and alone explains 94.8 % of within-battery capacity variance. It was also removed, giving Clean-6.
- The full classification, with code paths and formulas, is in `r3c3_feature_classification.md` and `r3c3_feature_classes.csv` (verified by `phase3_feature_audit.py`, 50/50).

4. C5 - Proxy-free improvements (Ch.4 sections 4.3-4.4)

Item	Result	Verdict
5.1 <code>ic_peak_V</code> (charge segment)	Clean-7 0.772 vs Clean-6 0.490 (15/20 folds)	Supported
5.2 Fixed discharge window 4.0-3.6 V	<code>win_dur</code> R2(cap) 0.995 vs <code>dis_duration</code> 0.972	Not established — window features are proxies
5.3 Self-referenced normalisation	0.810 -> 0.075; the B0018 reference is 6x smaller, which introduces a scale mismatch	Rejected
5.4 dSOH accumulation	0.810 -> -4.785; correlation of error with cycle position up to +0.993	Rejected
5.5 Window length 5-30 min	proxy-free flat at 0.842-0.843; proxy-laden rises 0.874 -> 0.914	No penalty

- The defensible headline is Clean-8 = 0.810 (Clean-6 + `ic_peak_V` + `t_40_41`), nested LOBO, 5 seeds. The paired claim on identical rows is 0.664 -> 0.810. The verify run notes that `t_40_41` alone is within noise (11/20 folds), so `ic_peak_V` is the supported addition; Ch.4 states this.
- Clean-6 = 0.490 is the strict proxy-free number. The 0.845/0.854 Proxy-Free arms contain a proxy-equivalent feature and are not quoted as proxy-free anywhere.

5. C6 - Wider validation

- 6.1 Learning curve, NASA B0025-B0056 (420 runs plus integrity checks): adding heterogeneous cells does not help. The pulsed-4A cells fail under every setting (B0027 R2 -132.8 at $k = 13$; Clean-8 mean -12.5, median +0.32), and widening the pool hurts same-condition tests (the room-2A protocol reproduces at 0.933 with $k = 3$ and falls to 0.63-0.85 with $k = 13$). Within-condition curves stay positive (medians -1.25 -> +0.15 -> +0.32 for $k = 3, 7, 13$). The claim is stated per condition. (Source: `phase4_learning_curve_summary.md`)
- 6.3 NASA -> CALCE CS2, four arms: frozen median R2 -37 (does not transfer); 1-cell scratch about -68; 1-cell warm-adapt about -31; within-CS2 control 0.39-0.68 on 6/8 cells. The failure is domain shift: a per-cell label offset (CS2 SOH spans 88-114 % vs NASA 72-108 %) and a charge rate of 0.5 C vs 0.75 C, which doubles the charge-side feature scale. The data and feature quality are sound; the control shows it. (Source: `phase4_cs2_summary.md`)

6. C7 - AE demoted to a documented negative result (Ch.4 section 4.6)

Evidence	Numbers
AE-on-TD vs raw TD	0.43 vs 0.85
AE on EIS vs PCA on EIS (gain over raw)	+0.08 vs +0.61
AE reconstruction	the interpolation grid, not the measurement (+0.1 mOhm offset)
E_fusion (EIS latent anchored by TD)	0.73, below TD-only 0.85

The E_fusion pipeline is positive on all four folds, including B0018 (+0.67), so the precise statement is: the EIS latent adds no value on top of a full TD feature set, and AE features are worse than raw features on both modalities. The AE as a representation is the negative result; a small gain on weak-feature regimes remains open.

APPROVED THESIS PHRASING (DOCUMENTED NEGATIVE RESULT)

"the autoencoder bottleneck adds no measurable value on either modality; its latent space reconstructs the interpolation grid rather than the impedance measurement; PCA is the stronger EIS reducer. The AE is reported as a negative result and the pipeline proceeds without it."

Recorded in [advisor_directives.md](#) (R3-C7 outcome)

7. C9 - Citation verification, all 13 studies

Full table: [r3c9_citation_verification.md](#). All 13 papers were downloaded and read from the full PDFs.

- **[11]** Sardar: random-split ET R2 0.9788 -> battery-wise 0.7855. Matches the audit.
- **[9]** Chen: B0006 LOCO R2 0.2464 -> 0.5286. Matches the audit (0.25 -> 0.53). Their protocol uses 3 cells, SOH = $Q(c)/Q(1)$, and the cycle index as an input; they note that their conformal intervals are not fully calibrated under severe domain shift. They also fail on B0006.
- **[10]** EWDC 0.975: a within-dataset LOBO (3 train, 1 held out; the same cells as ours: B0005/06/07/18 and CS2-35..38), average RMSE 1.21 %, MAE 0.65 %, 10 runs. The feature set is 17 charge-segment HIs including the cycle index and cumulative charged quantity, and the recursive predictor is anchored at $y_{hat_1} = 1$. The comparable number on our side is TD-All 0.927 (same protocol), not Clean-8.
- **[3]** uses $Q(k-1)$ as an input; **[4]** uses previous-cycle capacity and cycle number. Every strong unseen-battery claim in the list carries a proxy, which supports the Clean-8 framing.
- **[7]** is the only cross-battery result near R2 0.98: Oxford, 1-train/rest-test, per-cell capacity ratio. It is the smallest and cleanest scope in the list; the Ch.2 table annotates it.
- The B0018 fold of **[10]** has RMSE 1.91 % against their 1.21 % average, consistent with the B0018 behaviour in our own runs.

7b. Evidence on GitHub

All report, summary, verification and result files are in the private repository [DomIncham/soh-thesis-ae-bpnn](#):

What	File
Phase 1 - protocol fixes and proxy audit findings	phase1_oracle_findings.md
Phase 2 - two-axis experiment summary	phase2_summary_two_axes.md
Phase 3 - charge features (5.1)	phase3_charge_features_summary.md
Phase 3 - self-reference rejection (5.3)	phase3_selfref_summary.md
Phase 3 - window features (5.2/5.5)	phase3_window_summary.md
Phase 3 - dSOH rejection (5.4)	phase3_dsoh_summary.md
Phase 3 - window length (5.5)	phase3_windowlength_summary.md
Phase 4 - learning curve (6.1)	phase4_learning_curve_summary.md
Phase 4 - CALCE CS2 transfer (6.3)	phase4_cs2_summary.md
Phase 5 - AE negative result evidence (Round 2)	Progress_Report_Steps1-18.md , Steps_F3_I3_B7_Ablations.md
R3-C9 - 13-paper verification table	r3c9_citation_verification.md
Thesis drafts (v0.1)	Ch.1 , Ch.3 , Ch.4 , Ch.5 , Appendix
State card and experiment log	PROJECT_STATE.md

8. Verification status (re-run 2026-10-06)

Suite	Checks
phase1_verify	61/61 PASS
protocol_gap_verify	23/23 PASS
td_clean_verify	7/7 PASS
phase3_charge_verify	8/8 PASS
phase3_selfref_verify	9/9 PASS
phase3_window_verify	7/7 PASS
phase3_dsoh_verify	7/7 PASS
phase3_windowlength_verify	12/12 PASS
phase3_feature_audit	50/50 PASS
data_provenance_audit	26/26 PASS

Leakage probes: $\max|\text{diff}| = 0.00\text{e}+00$ on every protocol and clean feature set.

Two corrections came out of this re-verification. First, the Ch.4 draft had quoted the Oracle LOBO as 0.671; the correct phase-1 value is 0.663, and the draft was fixed. Second, the 0.845 and 0.854 Proxy-Free numbers come from different row sets, so Ch.4 now carries an explicit row-set note to prevent a cross-row-set comparison.

9. Draft status

- Ch.4 (Results), draft v0.1: [thesis_ch4_protocol_draft.md](#) - setup, protocol axis, proxy axis, the 5.1-5.5 verdicts, wider validation, the AE negative result, a headline table, and limitations.
- Ch.5 (Discussion), Ch.3 (Methodology), Ch.1 (Introduction) and the Appendix are drafted at v0.1; Ch.5 section 5.4 states the AE negative result with the E_fusion qualification.
- Ch.2 (Literature Review) is unblocked by C9. The comparison table can be built from [r3c9_citation_verification.md](#) with a unit and protocol note on each row.

ADVISOR DECISION REQUESTED — SECTION 10

Candidate Thesis Title Phrasing

Thesis title. A partial window is not proxy-free by construction, because window means are proxies at every length. I propose “proxy-audited, partial-window, unseen-battery SOH estimation”, which states the claim exactly, even though it is long. Please advise, or suggest a phrasing you prefer.

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